

PRD: Agent-Based Co-Scientist for Research-to-Device Synthesis

Absolutely — below is a strong first-draft PRD that sits **above** the two existing PRDs as a third product layer: an **agent-based co-scientist for research-to-device synthesis**, initially focused on **personal sound zones**.

Draft PRD v0.1 — Personal Sound Zone Co-Scientist

1. Product Summary

Build an **agent-based co-scientist** that operates on top of the existing Research Paper Knowledge Retrieval System and Reproducible Research Experimentation Environment to help researchers discover, evaluate, combine, and experimentally validate candidate approaches for applied device development.

The initial target application is a **personal sound zone device**: a system that creates localized audio regions where one listener can hear desired audio while nearby listeners experience reduced sound leakage, improved privacy, or reduced interference.

The co-scientist will convert research papers into structured **Approach Cards**, score them using evidence and device-relevance rubrics, generate candidate experiment plans, send approved experiments to the reproducible experimentation environment, interpret validation results, and synthesize candidate device architectures.

This product extends the existing retrieval PRD, which already defines PDF ingestion, structured chunking, Neo4j graph/vector retrieval, methods, datasets, metrics, claims, citations, evidence packs, and source-grounded answering. It also builds on the experimentation PRD, which already defines paper workspaces, reproducible experiment specs, containerized runtimes, AMD GPU support, FEM, Octave, Treble SDK workflows, artifact capture, validation reports, and reproduction packages.

The core product thesis is:

Convert a corpus of acoustics research into evidence-linked approach cards, score those approaches against device-relevant rubrics, generate reproducible experiments, update scores from validation results, and propose increasingly practical personal sound zone device architectures.

2. Problem Statement

Researchers and R&D teams often have access to large bodies of research but struggle to turn that literature into applied engineering direction.

For personal sound zones, relevant research may span:

- acoustic contrast control
- pressure matching
- beamforming
- null steering
- crosstalk cancellation
- active noise control
- adaptive filtering
- perceptual audio metrics
- room acoustics simulation
- head-tracking compensation
- microphone/speaker array calibration
- FEM or wave-based acoustic modeling
- embedded real-time DSP constraints

The existing retrieval system can answer what the literature says. The experimentation environment can reproduce and extend paper results. But there is still a missing layer:

1	Research evidence -> candidate approach -> experiment -> validation ->
2	device concept

Today, this synthesis is mostly manual. Researchers must read papers, compare incompatible metrics, infer hardware assumptions, decide what is worth testing, create experiment specs, run simulations, interpret results, and manually combine promising techniques into device concepts.

This product provides that missing synthesis layer.

3. Product Goals

The co-scientist should:

1. Convert research papers, claims, methods, metrics, and evidence into structured **Approach Cards**.
2. Add a domain-specific semantic layer for personal sound zone research.

3. Score candidate approaches using transparent, evidence-linked rubrics.
 4. Identify promising combinations of approaches for device development.
 5. Generate reproducible experiment specs that can be executed in the experimentation environment.
 6. Use experiment results to update approach scores and device concept confidence.
 7. Propose candidate personal sound zone device architectures.
 8. Preserve provenance from every recommendation back to source papers, claims, chunks, experiments, runs, and validation results.
 9. Distinguish between direct evidence, inferred synthesis, speculative hypotheses, and experimentally validated results.
 10. Keep a human researcher in the loop for expensive runs, external SDK calls, hardware tests, and device roadmap decisions.
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4. Non-Goals for MVP

The MVP should not attempt to:

1. Fully automate scientific discovery without researcher oversight.
 2. Build or control physical hardware directly.
 3. Guarantee that generated device concepts are patentable, manufacturable, or commercially viable.
 4. Replace expert acoustic engineering judgment.
 5. Reproduce arbitrary papers without human-defined assumptions and validation criteria.
 6. Autonomously run expensive GPU, FEM, Treble, or hardware experiments without approval.
 7. Perform full multimodal reasoning over all figures, schematics, and equations.
 8. Perform comprehensive IP, freedom-to-operate, safety, or regulatory analysis.
 9. Optimize embedded firmware or production DSP code.
 10. Support every scientific domain at launch.
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5. Target Users

5.1 Primary Users

Acoustics R&D researcher

Needs to discover, compare, and experimentally validate candidate methods for personal sound zone creation.

Research engineer

Needs to turn promising papers into reproducible simulations, parameter sweeps, and bench-test plans.

Device/product engineer

Needs to understand which research approaches can plausibly become a real device architecture.

Scientific ML / simulation researcher

Needs to test PINNs, FEM, Treble simulations, constrained optimization, or adaptive control strategies.

5.2 Secondary Users

Lab/platform administrator

Needs control over compute usage, credentials, approved runtimes, reproducibility, and audit logs.

Product owner / technical lead

Needs a ranked roadmap of candidate approaches, risks, validation gaps, and next experiments.

Knowledge manager

Needs to curate approach cards, merge duplicate concepts, correct entity extraction, and maintain the research graph.

6. Core User Stories

ID	User Story	Priority
CS-001	As a researcher, I want the system to convert retrieved papers into structured Approach Cards so that I can compare methods without reading every paper end to end.	P0
CS-002	As a researcher, I want each Approach Card to include evidence, assumptions,	P0

	metrics, limitations, hardware requirements, and citations.	
CS-003	As a researcher, I want the system to score approaches using a transparent rubric so that I can see which ideas are promising for personal sound zone devices.	P0
CS-004	As a researcher, I want scores to include confidence and supporting evidence so that I can distinguish strong evidence from weak inference.	P0
CS-005	As a research engineer, I want the system to generate experiment specs from Approach Cards so that I can test promising approaches reproducibly.	P0
CS-006	As a research engineer, I want to approve, edit, or reject generated experiment specs before they are executed.	P0
CS-007	As a researcher, I want validation results to update the approach scores so that the system learns from experiments.	P0
CS-008	As a device engineer, I want the system to synthesize candidate device concepts from validated approaches.	P1
CS-009	As a product owner, I want to see a ranked research	P1

	roadmap based on evidence, feasibility, risk, and information gain.	
CS-010	As an administrator, I want controls for compute, Treble SDK usage, credentials, and experiment approval.	P1
CS-011	As a researcher, I want the system to identify missing evidence and propose next-best experiments.	P1
CS-012	As a researcher, I want to compare multiple candidate device concepts side by side.	P2

7. Product Scope

7.1 MVP Scope

The MVP should include:

1. Personal sound zone semantic ontology.
2. Approach Card generation from retrieval evidence.
3. Evidence-linked scoring rubric.
4. Experiment-readiness scoring.
5. Experiment spec generation for simulation-first workflows.
6. Human approval workflow for generated experiments.
7. Integration with the reproducible experimentation environment.
8. Result ingestion from completed experiment runs.
9. Score updates based on validation results.
10. Candidate device concept generation.
11. Research roadmap view.
12. Traceability from recommendation to paper evidence and experiment artifacts.

7.2 Future Scope

Later releases may include:

1. Multi-agent autonomous experiment planning.
 2. Bench hardware integration.
 3. Closed-loop acoustic measurement with microphones and speaker arrays.
 4. Real-time DSP prototyping.
 5. Automated transfer-function measurement workflows.
 6. Embedded hardware feasibility estimation.
 7. BOM and manufacturability scoring.
 8. IP landscape analysis.
 9. Regulatory and safety review workflows.
 10. Multi-domain expansion beyond acoustics.
 11. Automated paper-code bootstrapping.
 12. Bayesian optimization over device and algorithm parameters.
 13. Multi-objective Pareto frontier visualization.
 14. Human subject perceptual testing workflows.
-

8. System Architecture

8.1 Reference Architecture

```
1 Layer 4: Co-Scientist Agent Layer
2 - Research Scout Agent
3 - Semantic Extractor Agent
4 - Hypothesis Generator Agent
5 - Experiment Designer Agent
6 - Simulation Runner Agent
7 - Rubric Critic Agent
8 - Device Integrator Agent
9 - Research Program Manager Agent
10
11 Layer 3: Scientific Semantic Layer
12 - Personal Sound Zone Ontology
13 - Approach Cards
14 - Experiment Cards
15 - Hypothesis Cards
16 - Device Concept Cards
17 - Rubric Scores
18 - Risk Registers
19 - Evidence Links
20 - Validation Links
21
22 Layer 2: Systems of Record
23 - Research Paper Knowledge Retrieval System
24 - Reproducible Research Experimentation Environment
25
26 Layer 1: Tools and Assets
```

```
27 - PDFs
28 - Paper chunks
29 - Neo4j knowledge graph
30 - Vector indexes
31 - Experiment specs
32 - Runtime images
33 - Treble SDK
34 - FEM solvers
35 - PINN/scientific ML runtimes
36 - Artifacts
37 - Metrics
38 - Validation reports
39
```

8.2 Workflow

```
1 1. User defines applied research goal
2   Example: "Create a device that produces a personal sound zone for
3   one listener."
4
5 2. Research Scout retrieves relevant papers and evidence.
6
7 3. Semantic Extractor converts papers into Approach Cards.
8
9 4. Rubric Critic scores approaches.
10
11 5. Hypothesis Generator proposes combinations of approaches.
12
13 6. Experiment Designer generates reproducible experiment specs.
14
15 7. Human approves, edits, or rejects experiment specs.
16
17 8. Experiment Runner executes approved experiments.
18
19 9. Validation results are captured.
20
21 10. Rubric Critic updates approach scores.
22
23 11. Device Integrator proposes candidate device architectures.
24
25 12. Research Program Manager recommends next experiments.
```

9. Semantic Layer Requirements

The semantic layer should create a typed, inspectable representation of research concepts that can be used for device synthesis.

9.1 Core Semantic Object Types

Object	Description
AcousticGoal	Desired sound-zone outcome, such as bright-zone fidelity, dark-zone attenuation, speech privacy, or leakage reduction.

ControlMethod	Algorithmic method, such as acoustic contrast control, pressure matching, beamforming, ANC, or adaptive filtering.
PhysicalScene	Assumptions about room, listener location, reverberation, source placement, receiver placement, and movement.
HardwareConcept	Speaker, microphone, sensor, amplifier, compute, and form-factor assumptions.
Metric	Evaluation metric, such as acoustic contrast, dark-zone attenuation, bright-zone error, latency, robustness, or calibration time.
ApproachCard	Structured representation of a research approach extracted from papers and evidence.
HypothesisCard	Proposed combination or extension of one or more approaches.
ExperimentCard	Structured experiment proposal linked to an approach or hypothesis.
ValidationResult	Result from an executed experiment, including pass/fail, metrics, plots, and artifacts.
DeviceConceptCard	Candidate applied device architecture assembled from validated approaches.
RiskRegister	Known risks, unresolved assumptions, failure modes, and mitigation ideas.

10. Approach Card

The **Approach Card** is the core product object. It converts research into an applied engineering candidate.

10.1 Approach Card Schema

```

1 approach_card:
2   id: approach_acc_nearfield_001
3   name: Acoustic contrast control for near-field personal sound zone

```

```
4 domain: personal_sound_zones
5
6 problem_fit:
7   application: personal_audio_privacy
8   target_goal:
9     - maximize_bright_zone_fidelity
10    - minimize_dark_zone_leakage
11    - preserve_speech_intelligibility
12   use_case:
13     - desktop_audio
14     - headrest_audio
15     - single_listener_private_zone
16
17 mechanism:
18   method_family: acoustic_contrast_control
19   summary: >
20     Optimizes loudspeaker drive signals to increase acoustic energy
21     in a target bright zone relative to a dark zone.
22   objective_function: evidence_linked_or_inferred
23   related_methods:
24     - pressure_matching
25     - beamforming
26     - adaptive_filtering
27
28 evidence:
29   source_papers: []
30   source_claims: []
31   supporting_chunks: []
32   tables_or_figures: []
33   reproduced_experiments: []
34   validation_results: []
35
36 reported_metrics:
37   - acoustic_contrast_db
38   - bright_zone_error
39   - dark_zone_spl
40   - robustness_to_position_shift
41
42 assumptions:
43   - listener_position_known_or_estimated
44   - room_transfer_functions_measurable
45   - speaker_array_has_sufficient_degrees_of_freedom
46   - calibration_possible
47
48 hardware_requirements:
49   speakers:
50     count_range: "4-16"
51     geometry: "array or distributed"
52   microphones:
53     required_for_calibration: true
54     required_for_runtime_feedback: optional
55   sensors:
56     head_tracking: optional
57   compute:
58     realtime_dsp_required: true
59
60 device_relevance:
61   possible_form_factors:
62     - desktop_bar
63     - headrest
64     - monitor_speaker_array
65     - chair_integrated_array
66   maturity: early_research_to_simulation
67
68 risks:
69   - sensitive_to_listener_movement
70   - may_require_recalibration
```

```

71     - low_frequency_control_limited_by_array_size
72     - room_variation_may_reduce_performance
73
74     next_experiments:
75         - compare_against_pressure_matching
76         - simulate_listener_shift
77         - test_under_reverberant_conditions
78         - measure_compute_latency
79

```

10.2 Approach Card Requirements

Requirement	Description	Priority
SEM-001	System creates Approach Cards from retrieved papers, chunks, methods, claims, and metrics.	P0
SEM-002	Each Approach Card must include source evidence links.	P0
SEM-003	Each Approach Card must distinguish direct evidence from inferred synthesis.	P0
SEM-004	Each Approach Card must include assumptions, hardware requirements, and failure modes.	P0
SEM-005	Each Approach Card must include relevant metrics and measurement definitions where available.	P0
SEM-006	Users can edit, approve, merge, or reject Approach Cards.	P1
SEM-007	System supports duplicate detection and alias resolution across approaches.	P1
SEM-008	System can mark an Approach Card as unscored, scored,	P1

	experimentally tested, validated, refuted, or superseded.	
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11. Personal Sound Zone Ontology

The MVP ontology should focus on device-relevant acoustics.

11.1 Acoustic Goals

```
1 bright_zone_fidelity
2 dark_zone_attenuation
3 acoustic_contrast
4 speech_privacy
5 listener_intelligibility
6 low_leakage_music_playback
7 multi_listener_isolation
8 robustness_to_movement
9 low_latency_playback
10 low_calibration_burden
11
```

11.2 Method Families

```
1 acoustic_contrast_control
2 pressure_matching
3 beamforming
4 null_steering
5 crosstalk_cancellation
6 active_noise_control
7 adaptive_filtering
8 modal_control
9 wave_field_synthesis
10 perceptual_weighting
11 head_tracking_compensation
12 transfer_function_estimation
13 room_impulse_response_modeling
14
```

11.3 Hardware Concepts

```
1 speaker_array
2 near_field_speaker_array
3 headrest_speaker_array
4 desktop_soundbar_array
5 distributed_loudspeakers
6 calibration_microphones
7 feedback_microphones
8 head_tracking_sensor
9 embedded_dsp
10 gpu_or_cpu_simulation_backend
11 amplifier_channel_array
12
```

11.4 Metrics

1	acoustic_contrast_db
2	bright_zone_spl_error
3	dark_zone_spl
4	dark_zone_attenuation_db
5	speech_transmission_index
6	speech_privacy_index
7	mse_pressure_error
8	frequency_response_error
9	latency_ms
10	compute_load
11	calibration_time
12	robustness_to_position_shift
13	room_generalization_score
14	power_consumption
15	speaker_count
16	microphone_count
17	

12. Agent Requirements

12.1 Agent Roles

Agent	Responsibility	MVP Priority
Research Scout Agent	Queries the retrieval system for relevant papers, methods, claims, and evidence.	P0
Semantic Extractor Agent	Converts evidence into Approach Cards.	P0
Rubric Critic Agent	Scores approaches and experiments using transparent rubrics.	P0
Experiment Designer Agent	Generates experiment specs from Approach Cards.	P0
Simulation Runner Agent	Submits approved experiments to the experimentation environment.	P1
Hypothesis Generator Agent	Proposes combinations of approaches.	P1
Device Integrator Agent	Synthesizes candidate device architectures.	P1

Research Program Manager Agent	Recommends the next highest-value experiments.	P2
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12.2 Agent Control Requirements

Requirement	Description	Priority
AGT-001	Agents must operate only on authorized corpora, experiments, and artifacts.	P0
AGT-002	Agents must provide evidence links for all claims and recommendations.	P0
AGT-003	Agents must mark unsupported or speculative statements explicitly.	P0
AGT-004	Agents must request human approval before running experiments.	P0
AGT-005	Agents must request human approval before using external paid services such as Treble.	P0
AGT-006	Agents must log all generated cards, scores, experiment specs, and decisions.	P0
AGT-007	Agents must preserve provenance back to papers, chunks, claims, experiments, runs, and artifacts.	P0
AGT-008	Agents must support critique/revision loops before final recommendations.	P1
AGT-009	Agents must not autonomously modify the	P1

corpus or delete artifacts.

13. Scoring Rubrics

The product needs two primary rubrics:

- 1. **Candidate Approach Rubric**
- 2. **Experiment Rubric**

Each score should include:

```
1 score: 0-5
2 confidence: low | medium | high
3 evidence_type: direct | inferred | experimental | user_supplied
4 evidence_links: []
5 rationale: short explanation
6
```

13.1 Candidate Approach Rubric

Dimension	Question	Weight
Evidence strength	Is the approach supported by papers, experiments, theory, or reproduced results?	15%
Reproducibility	Can the result be reproduced with available equations, code, data, or simulation details?	12%
Acoustic performance potential	Could the method produce meaningful bright-zone/dark-zone separation?	15%
Robustness	Does it survive listener movement, room variation, and imperfect calibration?	12%
Real-time feasibility	Can it run with acceptable latency on plausible hardware?	10%
Hardware feasibility	Can the required speaker/mic/sensor setup fit	10%

	into a plausible product?	
Calibration burden	Is setup practical for a user or installer?	8%
Composability	Can it combine with other methods?	8%
Measurement clarity	Are metrics well-defined and testable?	5%
Device relevance	Does it map to a personal sound zone product?	5%
Risk penalty	Does the approach have major safety, cost, manufacturability, usability, or IP risks?	subtractive

13.2 Candidate Approach Score

```

1 Approach Score =
2   0.15 * evidence_strength
3 + 0.12 * reproducibility
4 + 0.15 * acoustic_performance_potential
5 + 0.12 * robustness
6 + 0.10 * real_time_feasibility
7 + 0.10 * hardware_feasibility
8 + 0.08 * calibration_simplicity
9 + 0.08 * composability
10 + 0.05 * measurement_clarity
11 + 0.05 * device_relevance
12 - risk_penalty
13

```

13.3 Experiment Rubric

Dimension	Question	Weight
Hypothesis clarity	Does the experiment test one clear claim?	12%
Device relevance	Does it reduce uncertainty for the device?	15%
Baseline quality	Is there a fair comparison method?	10%

Metric quality	Are success/failure criteria explicit?	12%
Reproducibility	Are seed, code, container, data, hardware, and parameters captured?	12%
Information gain	Will the result change what we build next?	15%
Cost/time	Is the experiment cheap enough relative to expected learning?	8%
Failure informativeness	If it fails, will we learn why?	6%
Robustness coverage	Does it test movement, room variation, calibration error, or noise?	7%
Artifact quality	Will it produce reusable plots, impulse responses, metrics, reports, or code?	3%

14. Experiment Generation Requirements

14.1 Experiment Card Schema

```

1 experiment_card:
2   id: exp_psz_acc_vs_pm_001
3   approach_ids:
4     - approach_acc_nearfield_001
5     - approach_pressure_matching_001
6
7   objective: >
8     Compare acoustic contrast control against pressure matching
9     for a simulated desktop personal sound zone.
10
11   hypothesis: >
12     Acoustic contrast control will improve dark-zone attenuation
13     relative to pressure matching under fixed listener geometry,
14     but may be less robust under listener displacement.
15
16   baseline_methods:
17     - pressure_matching
18     - delay_and_sum_beamforming
19
20   independent_variables:
21     speaker_count: [4, 8, 12]

```

```

22 listener_shift_cm: [0, 5, 10, 20]
23 reverberation_condition: [anechoic, low, medium]
24 frequency_band_hz:
25     - [300, 1000]
26     - [1000, 4000]
27     - [4000, 8000]
28
29 fixed_assumptions:
30     room_geometry: desktop_small_room
31     bright_zone_position: listener_head_center
32     dark_zone_position: adjacent_listener
33     source_signal: speech_and_music_test_set
34
35 metrics:
36     - acoustic_contrast_db
37     - bright_zone_error
38     - dark_zone_spl
39     - robustness_to_position_shift
40     - latency_ms
41     - compute_load
42
43 validation:
44     pass_conditions:
45         acoustic_contrast_db_min: 12
46         bright_zone_error_max: 0.10
47         latency_ms_max: 20
48     comparison:
49         baseline_improvement_required: true
50
51 runtime:
52     preferred: python_numerics_or_treble
53     alternatives:
54         - dolfinx
55         - elmer
56         - octave
57
58 artifacts:
59     - transfer_functions
60     - impulse_responses
61     - metrics_json
62     - plots
63     - validation_report
64     - reproduction_manifest
65
66 approval:
67     requires_human_approval: true
68     estimated_cost: low
69     estimated_runtime: medium
70

```

14.2 Experiment Requirements

Requirement	Description	Priority
EXP-001	System generates experiment cards from Approach Cards.	P0
EXP-002	System generates experiment specs compatible with the experimentation environment.	P0

EXP-003	Experiment specs must include hypothesis, baseline, variables, metrics, validation criteria, runtime, and artifacts.	P0
EXP-004	System must support human approval before execution.	P0
EXP-005	System must estimate experiment cost, runtime, and required compute where possible.	P1
EXP-006	System must mark experiments as proposed, approved, running, completed, failed, or superseded.	P1
EXP-007	System must ingest validation results and update linked Approach Cards.	P1
EXP-008	System must support parameter sweeps for speaker count, listener shift, room conditions, and frequency bands.	P1

15. Candidate Device Concept

The **Device Concept Card** is the applied output of the co-scientist.

15.1 Device Concept Card Schema

```

1 device_concept_card:
2   id: device_psz_desktop_array_001
3   name: Near-field desktop personal sound zone bar
4
5   use_case:
6     primary: single_listener_private_desktop_audio
7     secondary:
8       - speech_privacy
9       - reduced_neighbor_leakage
10      - focused_music_playback

```

```

11
12 form_factor:
13     type: desktop_speaker_array
14     placement: under_monitor_or_desktop
15     listener_distance_cm: 50-100
16
17 acoustic_architecture:
18     control_stack:
19         - pressure_matching_baseline
20         - acoustic_contrast_control
21         - adaptive_filter_update_optional
22     calibration:
23         - measured_transfer_functions
24         - calibration_microphone
25         - optional_head_tracking
26     simulation_backing:
27         - room_impulse_response_modeling
28         - parameter_sweep_results
29
30 hardware:
31     speakers:
32         estimated_count: 8
33         geometry: linear_or_curved_array
34     microphones:
35         calibration_count: 1-2
36         runtime_feedback: optional
37     compute:
38         prototype: laptop_or_embedded_linux
39         production_candidate: embedded_dsp_or_edge_soc
40
41 expected_performance:
42     bright_zone: provisional
43     dark_zone: provisional
44     latency: unknown_until_bench_test
45     robustness: medium_risk
46
47 supporting_approaches:
48     - approach_acc_nearfield_001
49     - approach_pressure_matching_001
50     - approach_head_tracking_adaptive_001
51
52 supporting_experiments:
53     - exp_psz_acc_vs_pm_001
54     - exp_psz_listener_shift_001
55
56 unresolved_risks:
57     - low_frequency_leakage
58     - sensitivity_to_head_movement
59     - room_recalibration_burden
60     - embedded_compute_latency
61     - product_form_factor_constraints
62
63 next_steps:
64     - build_software_simulation_bench
65     - run_measured_transfer_function_experiment
66     - prototype_8_speaker_array
67     - test_listener_shift_robustness
68

```

15.2 Device Concept Requirements

Requirement	Description	Priority
-------------	-------------	----------

DEV-001	System generates candidate Device Concept Cards from scored Approach Cards.	P1
DEV-002	Each Device Concept Card must include supporting evidence and experiments.	P1
DEV-003	Each Device Concept Card must include unresolved risks and next experiments.	P1
DEV-004	System supports side-by-side comparison of device concepts.	P2
DEV-005	System can generate a prototype roadmap for a selected device concept.	P2
DEV-006	System can export Device Concept Cards as markdown, JSON, or report format.	P2

16. Data Model Extensions

The existing retrieval data model includes paper, author, section, chunk, table, figure, concept, method, dataset, metric, claim, and reference entities. The experimentation data model includes paper, experiment, run, artifact, and validation result entities.

This PRD adds the following semantic-layer entities.

16.1 New Node Types

Node	Description	Key Properties
ApproachCard	Research approach converted into applied candidate	approach_id, name, method_family, summary, status, maturity

HypothesisCard	Proposed combination or extension of approaches	hypothesis_id, text, rationale, status
ExperimentCard	Proposed experiment before execution	experiment_card_id, objective, hypothesis, runtime, approval_status
RubricScore	Score for approach, experiment, or device	score_id, rubric_type, dimension, score, confidence, rationale
Assumption	Explicit assumption behind an approach or experiment	assumption_id, text, confidence, status
FailureMode	Known or predicted way an approach can fail	failure_mode_id, description, severity, mitigation
HardwareComponent	Speaker, microphone, sensor, compute, or amplifier component	component_id, type, requirements, constraints
PhysicalScene	Room/listener/source/receiver setup	scene_id, geometry, reverberation, positions
DeviceConceptCard	Candidate applied device architecture	device_id, name, form_factor, status, maturity
ResearchRoadmapItem	Recommended next research action	roadmap_id, priority, expected_value, cost, status

16.2 New Relationships

Relationship	From → To	Description
DERIVED_FROM	ApproachCard → Claim/Chunk/Paper	Approach is derived from source evidence

USES_METHOD	ApproachCard → Method	Approach uses a research method
OPTIMIZES_FOR	ApproachCard → AcousticGoal	Approach targets an acoustic goal
MEASURED_BY	ApproachCard → Metric	Approach is evaluated by metric
REQUIRES_HARDWARE	ApproachCard → HardwareComponent	Approach requires hardware
ASSUMES	ApproachCard/ExperimentCard → Assumption	Assumption required for validity
FAILS_UNDER	ApproachCard → FailureMode	Known or predicted failure condition
SCORED_BY	ApproachCard/ExperimentCard/DeviceConceptCard → RubricScore	Object has rubric score
PROPOSES_EXPERIMENT	ApproachCard/HypothesisCard → ExperimentCard	Approach suggests experiment
EXECUTED_AS	ExperimentCard → Experiment/Run	Proposed experiment was executed
PRODUCED_RESULT	Run → ValidationResult	Run produced validation result
UPDATES_SCORE	ValidationResult → RubricScore	Result changes confidence or score
COMBINES_APPROACH	DeviceConceptCard → ApproachCard	Device concept uses approach
HAS_NEXT_STEP	DeviceConceptCard → ResearchRoadmapItem	Device suggests next research action

17. API Requirements

17.1 Core APIs

Endpoint	Purpose	Priority
POST /co-scientist/goals	Create an applied research goal.	P0
POST /co-scientist/research-scout	Retrieve evidence relevant to a goal.	P0
POST /approach-cards/generate	Generate Approach Cards from evidence packs.	P0
GET /approach-cards/{id}	Retrieve an Approach Card.	P0
POST /approach-cards/{id}/score	Score or rescore an Approach Card.	P0
POST /experiment-cards/generate	Generate experiment proposals from Approach Cards.	P0
POST /experiment-cards/{id}/approve	Approve an experiment for execution.	P0
POST /experiment-cards/{id}/execute	Submit approved experiment to experimentation environment.	P1
POST /validation-results/ingest	Ingest result from experiment environment.	P1
POST /device-concepts/generate	Generate candidate device concepts.	P1
GET /roadmap	Retrieve prioritized research roadmap.	P1
POST /co-scientist/feedback	Submit user feedback on cards, scores, or recommendations.	P1

18. User Interface Requirements

18.1 Co-Scientist Workspace

Requirement	Description	Priority
UI-CS-001	User can define an applied research goal.	P0
UI-CS-002	User can view retrieved evidence packs.	P0
UI-CS-003	User can view, edit, approve, reject, and merge Approach Cards.	P0
UI-CS-004	User can inspect rubric scores and score rationales.	P0
UI-CS-005	User can see source citations for every claim and score.	P0
UI-CS-006	User can generate experiment proposals from selected Approach Cards.	P0
UI-CS-007	User can approve or edit generated experiment specs.	P0
UI-CS-008	User can view experiment status and validation results.	P1
UI-CS-009	User can compare candidate device concepts.	P1
UI-CS-010	User can view a research roadmap ranked by evidence, feasibility, risk, and expected information gain.	P1

19. Example User Journey

Journey A — From Literature to Personal Sound Zone Device Concept

1. User creates a goal:

```
1 | Create a device that forms a private sound zone for one listener at a
2 | desktop.
```

2. Research Scout queries the retrieval system for:

```
1 | personal sound zones
2 | acoustic contrast control
3 | pressure matching
4 | beamforming
5 | crosstalk cancellation
6 | adaptive sound zones
7 | room impulse response simulation
8 | speaker array optimization
9 |
```

3. Semantic Extractor creates Approach Cards.

4. Rubric Critic scores each approach.

5. User reviews the top candidates:

```
1 | Approach 1: Acoustic contrast control
2 | Approach 2: Pressure matching
3 | Approach 3: Adaptive filtering with head tracking
4 | Approach 4: Crosstalk cancellation
5 | Approach 5: Beamforming / null steering
6 |
```

6. Hypothesis Generator proposes:

```
1 | A near-field desktop speaker array using pressure matching as a
2 | baseline,
3 | acoustic contrast control for dark-zone reduction, and optional head
4 | tracking
5 | to improve robustness under listener movement.
```

7. Experiment Designer creates an experiment spec.

8. User approves the experiment.

9. Experiment Runner submits it to the reproducible environment.

10. Validation results update the Approach Cards.

11. Device Integrator proposes:

```
1 | Near-field desktop personal sound zone bar
2 |
```

12. Research Program Manager recommends next experiments:

```
1 | 1. Measure robustness to 5-20 cm listener movement.
2 | 2. Compare 4, 8, and 12 speaker configurations.
```

- 3 3. Test recalibration burden under room changes.
- 4 4. Evaluate low-frequency leakage.
- 5 5. Measure real-time compute latency.
- 6

20. Success Metrics

20.1 Product Metrics

Metric	Target
Approach Card generation usefulness	$\geq 75\%$ of generated Approach Cards rated useful by researchers
Evidence traceability	100% of Approach Cards include source links or are marked insufficient evidence
Score explainability	$\geq 90\%$ of rubric scores include rationale and evidence/confidence indicators
Experiment proposal usefulness	$\geq 70\%$ of generated experiments accepted or edited rather than discarded
Research time saved	Users report $\geq 40\%$ reduction in time to identify testable candidate approaches
Device concept usefulness	$\geq 60\%$ of generated device concepts rated directionally useful by product/research leads

20.2 Scientific Quality Metrics

Metric	Target
Evidence grounding	$\geq 90\%$ of generated approach claims supported by source evidence or marked inferred
Unsupported claim rate	$\leq 5\%$ in evaluated outputs
Approach deduplication precision	$\geq 85\%$ for known duplicate/alias method families
Experiment spec validity	$\geq 85\%$ of generated specs pass schema validation

Reproducibility metadata completeness	≥ 95% for executed experiments
Validation linkage	100% of completed runs linked back to Approach Cards or Experiment Cards

20.3 Device Research Metrics

Metric	Target
Initial approach coverage	At least 25 relevant personal sound zone Approach Cards created from the seed corpus
Experiment conversion	At least 10 Approach Cards converted into experiment proposals
First simulation loop	At least 3 generated experiments executed end to end
Candidate concept generation	At least 3 personal sound zone device concepts generated
Roadmap quality	Research lead can identify top 5 next experiments from system output

21. MVP Acceptance Criteria

The MVP is complete when:

1. A user can define a personal sound zone research goal.
2. The system can retrieve relevant evidence from the research paper retrieval system.
3. The system can generate Approach Cards from retrieved papers, chunks, methods, metrics, and claims.
4. Each Approach Card includes evidence links, assumptions, metrics, limitations, and hardware implications.
5. The system can score Approach Cards using the candidate approach rubric.
6. Scores include confidence, rationale, and evidence status.
7. The system can generate at least one reproducible experiment spec from an Approach Card.
8. The user can approve, edit, or reject the generated experiment spec.
9. An approved experiment can be sent to the experimentation environment.

10. A completed validation result can be ingested and linked back to the original Approach Card.
 11. The system can update approach scores based on validation results.
 12. The system can generate at least one candidate personal sound zone Device Concept Card.
 13. Every recommendation is traceable to source evidence, generated hypotheses, or validation results.
 14. The system can explicitly state when there is not enough evidence.
-

22. Recommended MVP Implementation Plan

Phase 1 — Semantic Layer Foundation

Deliverables:

1. Personal sound zone ontology.
2. Approach Card schema.
3. Rubric schema.
4. Neo4j schema extensions.
5. Evidence-to-card extraction prompt/tooling.
6. Manual review interface for generated Approach Cards.

Acceptance criteria:

```
1 | For a seed corpus of personal sound zone papers, the system can  
2 | generate  
3 | structured Approach Cards with citations, assumptions, metrics, and  
  | limitations.
```

Phase 2 — Scoring and Comparison

Deliverables:

1. Candidate approach rubric implementation.
2. Experiment-readiness rubric implementation.
3. Score rationale generation.
4. Evidence/confidence indicators.
5. Approach comparison table.
6. Pareto-style comparison view.

Acceptance criteria:

```
1 | A researcher can compare candidate approaches by evidence strength,
```

```
2 device feasibility, robustness, reproducibility, and risk.  
3
```

Phase 3 — Experiment Proposal Generation

Deliverables:

1. Experiment Card schema.
2. Experiment spec generator.
3. Validation criteria generator.
4. Human approval workflow.
5. Integration with experimentation environment schema.
6. Experiment proposal dashboard.

Acceptance criteria:

```
1 | The system can generate valid experiment specs for simulation-first  
2 | personal sound zone experiments.  
3 |
```

Phase 4 — Closed-Loop Validation

Deliverables:

1. Experiment execution handoff.
2. Run/result ingestion.
3. Validation result parser.
4. Score update mechanism.
5. Approach status transitions.
6. Experiment result dashboard.

Acceptance criteria:

```
1 | A completed experiment updates the linked Approach Card and changes  
2 | the research roadmap.  
3 |
```

Phase 5 — Device Concept Synthesis

Deliverables:

1. Device Concept Card schema.
2. Device Integrator Agent.

- 3. Candidate concept generation.
- 4. Risk register generation.
- 5. Next-experiment roadmap.
- 6. Device concept comparison view.

Acceptance criteria:

1	The system can synthesize at least three candidate personal sound zone device concepts from scored approaches and validation results.
2	
3	

23. Key Risks and Mitigations

Risk	Impact	Mitigation
Approach Cards overstate evidence	User trust loss	Require source links, confidence, and direct/inferred labels
Rubric scores feel arbitrary	Poor adoption	Make all scoring dimensions inspectable and editable
Generated experiments are invalid	Wasted researcher time	Use schema validation and human approval before execution
Agents hallucinate technical feasibility	Bad device decisions	Require citations or mark as speculative
Research papers use incompatible metrics	Poor comparison quality	Normalize metrics and preserve original reported metrics
Acoustic simulations do not match physical prototypes	False confidence	Require staged validation: simulation → measured transfer functions → bench prototype
Treble or GPU runs create unexpected cost	Budget risk	Require approval, cost estimate, and quota controls
Personal sound zone performance is highly context-	Overgeneralization	Track physical scene assumptions and robustness

dependent		conditions
Device concepts ignore manufacturability	Unrealistic roadmap	Include hardware feasibility and calibration burden in scoring
Knowledge graph becomes too complex	Operational burden	Start with a small schema and add node types only when needed

24. Open Product Decisions

1. What is the initial seed corpus for personal sound zone research?
2. Should the first MVP focus on desktop, headrest, monitor-mounted, or room-based personal sound zones?
3. What is the first target form factor?
4. What is the minimum acceptable acoustic contrast target?
5. What frequency range matters most for the first prototype?
6. Should the first experiments use simplified Python simulation, Treble, FEM, or measured transfer functions?
7. What is the baseline method: pressure matching, delay-and-sum beamforming, or another control method?
8. Should head tracking be in MVP or deferred?
9. What level of movement robustness is required: 5 cm, 10 cm, 20 cm, or more?
10. What is the acceptable calibration burden for a user?
11. Should experiment execution require explicit approval every time?
12. Should the co-scientist optimize for scientific novelty, device feasibility, or fastest path to prototype?
13. What score threshold allows an approach to become part of a Device Concept Card?
14. Should low-evidence/high-upside approaches be preserved in a separate exploratory lane?

25. Recommended MVP Cut

For the first usable release, keep the MVP focused on:

Must-have

1. Personal sound zone ontology.
2. Approach Card generation.
3. Evidence-linked source citations.
4. Candidate approach rubric.
5. Experiment-readiness rubric.
6. Experiment spec generation.
7. Human approval workflow.
8. Integration with the existing experiment environment.
9. Result ingestion.
10. Device Concept Card generation.

Defer

1. Fully autonomous agent loops.
2. Physical hardware control.
3. Real-time DSP implementation.
4. Automated microphone/speaker calibration.
5. Head-tracking integration.
6. IP analysis.
7. Manufacturability analysis.
8. Regulatory analysis.
9. Multi-domain support.
10. Public web-scale research discovery.

26. Strongest Next Step

The best next step is to select a **seed corpus of 20–30 personal sound zone papers** and use it to validate the semantic layer.

The first implementation milestone should be:

```
1 | Given a personal sound zone research goal and a corpus of relevant
2 | papers,
3 | generate 20+ Approach Cards with evidence links, score them using the
4 | candidate approach rubric, and produce the top 5 recommended
   | experiments.
```

That gets you from “retrieval system” to “research co-scientist” without prematurely building full autonomy or hardware integration.